

# **Week 11: Data Analysis and Interpretation Part I**

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# Housekeeping

# Project Timeline

Only 6 weeks left! ALL IRBs have been submitted

Goal is to launch survey as soon as possible (continue to build Qualtrics survey)

We will work with as many people as we can get!

After Paper #3, front end of study (lit review, RQs/Hs, should be complete)

# What's Next?

## Week 12 11/10 Advanced Analyses

This week will focus on complex data analytic techniques that go beyond basic ANOVA and regression methods. The lecture will include a general overview of several statistical

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methods that advance communication science by answering nuanced research questions and hypotheses: structural equation modeling, hierarchical linear modeling, meta-analyses, logistic regression, Bayesian modeling, measurement invariance, and growth curve modeling.

**Workshop Topic:** Descriptive Statistics and Reliability

We will practice generating descriptive statistics, creating composite measures, and exploring the reliability of our data. I will provide a dataset for you.

### Readings

\*Gliner, J. A., Morgan, G. A., & Leech, N. L. (2017). *Research methods in applied settings: An integrated approach to design and analysis* (3<sup>rd</sup> ed.). Routledge. \*Read Ch 22\*

Frey, T. K., & Vallade, J. I. (2023). Securing the right skills: A longitudinal assessment of college students' writing and public speaking self-efficacy. *Basic Communication Course Annual*, 35, 31-66. <https://ecommons.udayton.edu/bcca/vol35/iss1/6/>

Goodboy, A. K., & Kline, R. B. (2017). Statistical and practical concerns with published communication research featuring structural equation modeling. *Communication Research Reports*, 34(1), 68-77. <https://doi.org/10.1080/08824096.2016.1214121>

Walter, N., Cohen, J., Holbert, R. L., & Morag, Y. (2020). Fact-checking: A meta-analysis of what works and for whom. *Political Communication*, 37(3), 350-375.

<https://doi.org/10.1080/10584609.2019.1668894>

### Statistical Problem Set #1

# Overview of Today

## Variable Relationships!

- Brief Review
- Correlation
- Partial Correlation
- Multiple Regression

# Brief Review on BG and WS Difference Tests

Already Covered:

- Descriptive Statistics
- Inferential Statistics
- Nonparametric tests (Chi-Square)
- Group Differences (t-tests)
- ANOVA, Factorial ANOVA,
- ANCOVA, MANOVA, Repeated Measures MANOVA (Sort of)

# Visualizing

Group 1<sub>T1</sub>

Group 1<sub>population</sub>

Chi-Square  $\chi^2$  (16

# Independent Samples *t*-test (17)

Group 1<sub>T1</sub>

Group 2<sub>T1</sub>



# Paired-Samples *t*-test (17)

Group 1<sub>T1</sub>

Group 1<sub>T2</sub>

Group 1<sub>T1</sub>

Group 2<sub>T1</sub>

Group 3<sub>T1</sub>

One-way analysis of  
variance (ANOVA)  $F(18)$

Group 1<sub>T1</sub>

Group 4<sub>T1</sub>

Group 2<sub>T1</sub>

Group 5<sub>T1</sub>

Group 3<sub>T1</sub>

Two-way between-groups analysis of  
variance (Factorial ANOVA)  $F(19)$

Group 1<sub>T1</sub>

Group 1<sub>T2</sub>

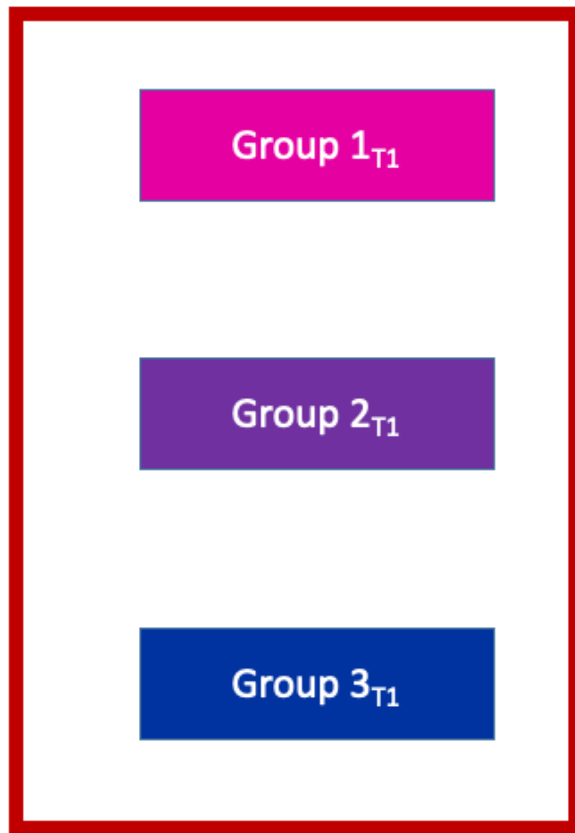
Group 2<sub>T1</sub>

Group 2<sub>T2</sub>

Group 3<sub>T1</sub>

Group 3<sub>T2</sub>

Mixed between-within subject  
analysis of variance (Repeated  
Measures ANOVA)  $F(20)$



Analysis of covariance (ANCOVA)  $F($

# Quick Knowledge Check

What is the difference between a One-way ANOVA and Factorial ANOVA?

# Real Life

**H1a-c:** Instructor syllabus policies with greater controlling language will produce a) increased threat to freedom, b) increased reactance, and c) decreased intent to comply than syllabus policies with less controlling language.

**H2a-c:** Instructor syllabus policies perceived as unfair will produce a) increased threat to freedom, b) increased reactance, and c) decreased intent to comply than syllabus policies perceived as fair.

# Real Life

Comparing Means

Threat to Freedom

Reactance

Compliance

2 x 2 Factorial Design

**Table 1.** Effects of controlling language and fairness on dependent measures.

| Variable          | Controlling Language | Fairness     |             | Significant Effects |
|-------------------|----------------------|--------------|-------------|---------------------|
|                   |                      | Fair         | Unfair      |                     |
| Threat to freedom | Low                  | 2.49 (0.95)  | 3.56 (0.94) | L*                  |
|                   | High                 | 3.27 (1.11)  | 3.71 (1.04) | F*                  |
| Reactance         | Low                  | -1.53 (1.49) | .96 (1.43)  | LF*                 |
|                   | High                 | -.37 (1.69)  | .86 (1.70)  | L*                  |
| Intent to comply  | Low                  | 5.27 (1.19)  | 4.83 (1.44) | F*                  |
|                   | High                 | 4.55 (1.27)  | 4.67 (1.48) | LF*                 |

L = main effect for controlling language, F = main effect for fairness, LF = two-way interaction; \*\*  $p < .10$ , \*  $p < .05$ .



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  - EX: Do gender and class time (morning, noon, night) interact to affect achievement?
- 1 between groups IV (4 levels), 1 normally-distributed DV
  - EX: Do NCAA basketball tournament regions differ in terms of field goal percentage?

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- A 2 x 3 x 2 mixed design with repeated measures on the third variable, 1 normally-distributed DV

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  - EX: Do gender and class time (morning, noon, night) lead students to have better grades between Fall and Spring semester?
- 1 between groups IV (4 levels), 3 normally-distributed DVs

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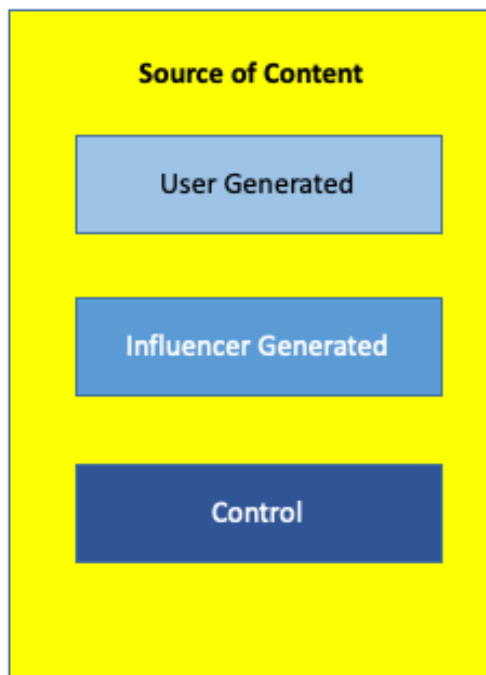
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- A 2 x 3 x 2 mixed design with repeated measures on the third variable, 1 normally-distributed DV
  - EX: Do gender and class time (morning, noon, night) lead students to have better grades between Fall and Spring semester?
- 1 between groups IV (4 levels), 3 normally-distributed DVs
  - EX: Does teacher type (lecturer, non-tenured professor, graduate student, tenured professor) differ in terms of their credibility (competence, caring, goodwill)?

# Example Study 1

Overview

Test?

## Team PvP



H1: User-generated content will be perceived as having more *goodwill* than influencer-sponsored content.

H2: User-generated content will be perceived as more *trustworthy* than influencer-sponsored content.

H3: Influencer-sponsored content will be perceived as more *competent* than user-generated content.



# Example Study 2

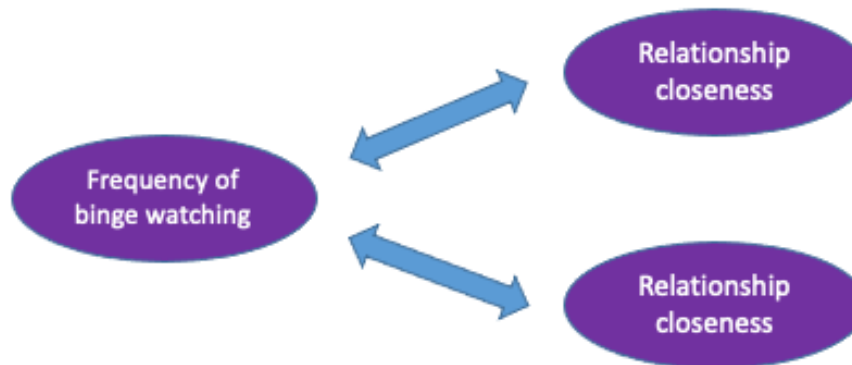
Overview

Option 1a

Option 1b

Option 2

## Team Binge



**H1:** Frequent binge-watching in romantic relationships will be *positively* associated with relationship closeness.

**H2:** Frequent binge-watching in romantic relationships will be associated with conflict regularity.



# Example Study 3

Overview

Test?

## Team Snowstorm



Confidence

**H1:** Participants who receive personalized, role-specific UK alerts will report greater confidence in their ability to interpret and take action than recipients of generic alerts.

Behavioral  
Intention

**H2:** Participants who receive personalized, role-specific UK alerts will report stronger behavioral intent to take action when compared to generic alerts.



# What If!?

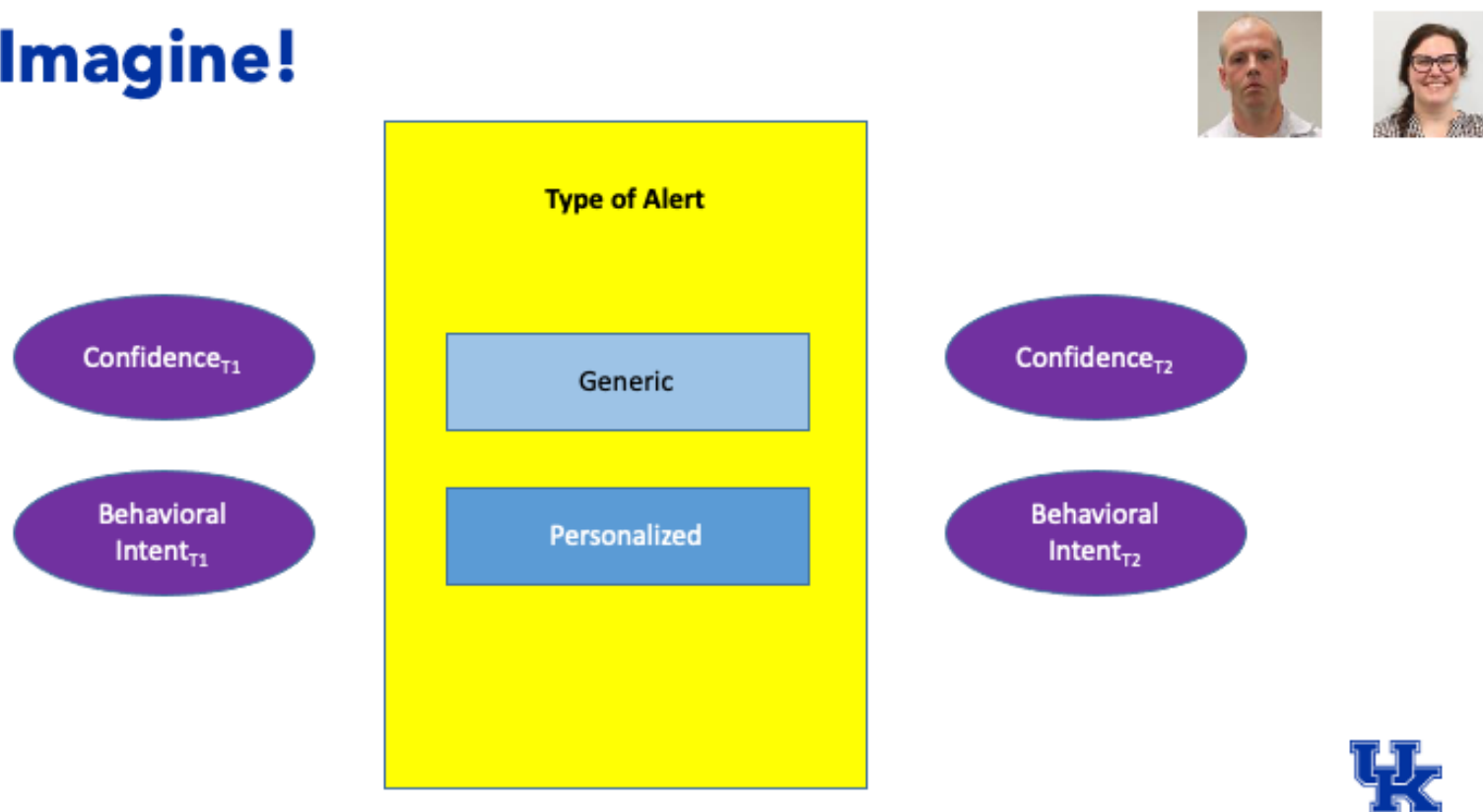
Overview

Option 1

Option 2

Option 3

**Imagine!**



# GLM Ch. 21 & (Parts of) 22

*21: Analysis and Interpretation of Basic Associational Research Questions*

*22: Analysis and Interpretation of Complex Associational Questions*

# Ch. 21 Overview

Chapter 21 (p. 369-385) covers:

- Analyzing Continuous Variables with Parametric Statistics
  - Pearson Product-Moment Correlation ( $r$ )
  - Statistical Significance
  - Correlation Matrix
  - Effect Sizes
  - Confidence Intervals
- Using Nonparametric Associational Statistics
  - Spearman Rank-Order Correlation Coefficient ( $\rho$ )
  - Kendall Tau Coefficient ( $\tau$ )
  - Misleading Correlation Coefficients (p. 374-376)
  - Associational Statistics for Nominal Variables
  - The Chi-Square Test and Effect Size Indices



# Remember Pearson

# What are correlations?

An index of the strength of the linear relationship between two variables

- Ranges from -1 to +1
- A positive relationship means that as scores on one variable increase, scores on the other variable also increase.
- An inverse relationship means that a high score on one variable is associated with a low score for the same person on the other variable and vice versa.

# Correlation Matrix

**Table 1** Means Table

| Measure                         | <i>M</i> | <i>SD</i> | Reliability | Range |
|---------------------------------|----------|-----------|-------------|-------|
| <i>Caring</i>                   |          |           |             |       |
| Teacher Self-Report             | 35.02    | 4.76      | .86         | 6–42  |
| Perceptions of Supervisor       | 24.15    | 9.81      | .94         | 6–42  |
| <i>Burnout</i>                  |          |           |             |       |
| Emotional Exhaustion            | 21.50    | 7.70      | .88         | 9–45  |
| Depersonalization               | 10.92    | 4.02      | .73         | 5–25  |
| Loss of Personal Accomplishment | 15.13    | 4.77      | .81         | 7–35  |
| <i>Temperament</i>              |          |           |             |       |
| Extraversion                    | 29.92    | 6.51      | .81         | 6–42  |
| Agreeableness                   | 42.87    | 8.05      | .87         | 8–56  |
| Conscientiousness               | 49.17    | 6.00      | .86         | 8–56  |
| Neuroticism                     | 22.67    | 7.83      | .81         | 7–49  |
| Openness to Experience          | 39.96    | 7.79      | .81         | 8–56  |
| <i>Organizational Outcomes</i>  |          |           |             |       |
| Teacher Motivation              | 23.41    | 3.64      | .92         | 4–28  |
| Teacher Job Satisfaction        | 28.07    | 6.91      | .96         | 5–35  |

**Table 2** Correlations Among Caring, Burnout, and Organizational Variables

|                                    | 2    | 3    | 4    | 5     | 6    | 7     |
|------------------------------------|------|------|------|-------|------|-------|
| 1. Caring (self-report)            | –.39 | –.56 | –.26 | .19*  | .30  | .04*  |
| 2. Emotional Exhaustion            | –    | –.76 | –.39 | –.16* | –.67 | –.50  |
| 3. Depersonalization               | –    | –    | .42  | –.11* | –.61 | –.33  |
| 4. Loss of Personal Accomplishment | –    | –    | –    | –.07* | –.15 | –.18* |
| 5. Supervisor Caring               | –    | –    | –    | –     | .17* | .38   |
| 6. Teacher Motivation              | –    | –    | –    | –     | –    | .64   |
| 7. Job Satisfaction                | –    | –    | –    | –     | –    | –     |

*Note:* All correlations are significant at the .05 level (at least) unless specified.

\*Not statistically significant.

Teven, J.J. (2007). Teacher temperament: Correlates with teacher caring, burnout, and organizational outcomes. *Communication Education*, 56, 382-400

# Correlation Matrix

Table 12

*Means, Standard Deviations, and Zero-Order Correlations for Variables in Phase 2 for the Full Sample (N = 803)*

| Variable                | Full Sample |      | 1      | 2      | 3      | 4      | 5      | 6      | 7      | 8      | 9      | 10     | 11     | 12     | 13    | 14     | 15     | 16    | 17    |
|-------------------------|-------------|------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-------|--------|--------|-------|-------|
|                         | M           | SD   |        |        |        |        |        |        |        |        |        |        |        |        |       |        |        |       |       |
| 1. Mastery Experience   | 4.4         | 1.1  |        |        |        |        |        |        |        |        |        |        |        |        |       |        |        |       |       |
| 2. Vicarious Experience | 3.9         | 1.2  | .51**  |        |        |        |        |        |        |        |        |        |        |        |       |        |        |       |       |
| 3. Social Persuasions   | 3.7         | 1.3  | .73**  | .61**  |        |        |        |        |        |        |        |        |        |        |       |        |        |       |       |
| 4. Physiological State  | 2.6         | 1.3  | -.63** | -.39** | -.51** |        |        |        |        |        |        |        |        |        |       |        |        |       |       |
| 5. Math Grade SE        | 4.9         | 1.0  | .77**  | .44**  | .61**  | -.55** |        |        |        |        |        |        |        |        |       |        |        |       |       |
| 6. Math Skills SE       | 80.8        | 15.3 | .62**  | .43**  | .52**  | -.44** | .62**  |        |        |        |        |        |        |        |       |        |        |       |       |
| 7. Math Courses SE      | 3.4         | 1.5  | .48**  | .32**  | .44**  | -.35** | .46**  | .51**  |        |        |        |        |        |        |       |        |        |       |       |
| 8. Self-Regulatory SE   | 4.4         | 1.1  | .62**  | .63**  | .58**  | -.57** | .60**  | .57**  | .42**  |        |        |        |        |        |       |        |        |       |       |
| 9. Self-Concept         | 4.4         | 1.0  | .88**  | .54**  | .73**  | -.65** | .73**  | .62**  | .49**  | .65**  |        |        |        |        |       |        |        |       |       |
| 10. Inviting Self       | 4.3         | 1.1  | .56**  | .62**  | .59**  | -.47** | .56**  | .49**  | .35**  | .72**  | .58**  |        |        |        |       |        |        |       |       |
| 11. Inviting Others     | 4.5         | 1.0  | .31**  | .46**  | .38**  | -.28** | .34**  | .33**  | .22**  | .51**  | .32**  | .60**  |        |        |       |        |        |       |       |
| 12. Task Goals          | 3.9         | 1.2  | .44**  | .72**  | .52**  | -.43** | .41**  | .39**  | .28**  | .63**  | .48**  | .61**  | .46**  |        |       |        |        |       |       |
| 13. Approach Goals      | 4.2         | 1.2  | .12**  | .34**  | .20**  | .02    | .08*   | .19**  | .10*   | .15**  | .13**  | .18**  | .15**  | .24**  |       |        |        |       |       |
| 14. Avoidance Goals     | 3.1         | 1.2  | -.23** | .04    | -.11*  | .37**  | -.23** | -.16** | -.13** | -.18** | -.22** | -.10*  | -.03   | -.02   | .45** |        |        |       |       |
| 15. Self-Handicapping   | 2.6         | 1.2  | -.28** | -.18** | -.16** | .31**  | -.25** | -.23** | -.20** | -.32** | -.28** | -.21** | -.22** | -.18** | .00   | .25**  |        |       |       |
| 16. Optimism            | 4.2         | 0.9  | .48**  | .46**  | .47**  | -.46** | .45**  | .43**  | .31**  | .53**  | .49**  | .53**  | .37**  | .48**  | .06   | -.25** | -.23** |       |       |
| 17. Teacher Rating      | 7.3         | 1.9  | .46**  | .06    | .31**  | -.28** | .42**  | .44**  | .30**  | .21**  | .42**  | .14**  | .16**  | .05    | .04   | -.24** | -.20** | .19** |       |
| 18. Semester GPA        | 85.4        | 8.2  | .63**  | .16**  | .40**  | -.34** | .59**  | .45**  | .33**  | .33**  | .56**  | .26**  | .24**  | .14**  | .01   | -.23** | -.28** | .26** | .69** |

Note. For calculations with Semester GPA,  $N = 784$ . SE = self-efficacy.

\*  $p < .05$ . \*\*  $p < .001$ .

Usher, E. L. (2007). Tracing the origins of confidence: A mixed methods exploration of the sources of self-efficacy beliefs in mathematics (Unpublished doctoral dissertation). Emory University, Atlanta, GA.

# Correlation Matrix

Table 13

*Means, Standard Deviations, and Zero-Order Correlations for Variables in Phase 2 by Gender*

| Variable                | Girls    |           |        |        |        |        |        |        |        |        |        |        |        |        |       |        |        |        |        |        | Boys     |           |
|-------------------------|----------|-----------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-------|--------|--------|--------|--------|--------|----------|-----------|
|                         | <i>M</i> | <i>SD</i> | 1      | 2      | 3      | 4      | 5      | 6      | 7      | 8      | 9      | 10     | 11     | 12     | 13    | 14     | 15     | 16     | 17     | 18     | <i>M</i> | <i>SD</i> |
| 1. Mastery Experience   | 4.3      | 1.2       |        | .51**  | .71**  | -.61** | .78**  | .64**  | .48**  | .59**  | .88**  | .58**  | .34**  | .45**  | .20** | -.13*  | -.27** | .55**  | .39**  | .57**  | 4.4      | 1.1       |
| 2. Vicarious Experience | 3.9      | 1.2       | .53**  |        | .64**  | -.38** | .42**  | .43**  | .27**  | .58**  | .54**  | .62**  | .48**  | .72**  | .37** | .11*   | -.15*  | .50**  | .03    | .16*   | 3.9      | 1.2       |
| 3. Social Persuasions   | 3.7      | 1.3       | .75**  | .58**  |        | -.47** | .58**  | .50**  | .41**  | .56**  | .72**  | .61**  | .42**  | .55**  | .28** | .00    | -.13*  | .50**  | .23**  | .35**  | 3.8      | 1.3       |
| 4. Physiological State  | 2.7      | 1.3       | -.65** | -.40** | -.54** |        | -.50** | -.41** | -.30** | -.51** | -.62** | -.46** | -.29** | -.43** | -.08  | .29**  | .30**  | -.46** | -.17** | -.28** | 2.5      | 1.2       |
| 5. Math Grade SE        | 4.9      | 1.0       | .76**  | .46**  | .64**  | -.59** |        | .65**  | .47**  | .58**  | .75**  | .57**  | .36**  | .39**  | .12*  | -.17** | -.22** | .48**  | .39**  | .58**  | 4.9      | 1.0       |
| 6. Math Skills SE       | 79.4     | 15.8      | .61**  | .43**  | .54**  | -.46** | .60**  |        | .51**  | .55**  | .64**  | .51**  | .35**  | .39**  | .25** | -.09   | -.19** | .43**  | .47**  | .50**  | 82.2     | 14.6      |
| 7. Math Courses SE      | 3.4      | 1.5       | .48**  | .37**  | .47**  | -.40** | .45**  | .51**  |        | .38**  | .47**  | .35**  | .24**  | .25**  | .10*  | -.13*  | -.23** | .28**  | .32**  | .32**  | 3.4      | 1.4       |
| 8. Self-Regulatory SE   | 4.5      | 1.1       | .65**  | .68**  | .60**  | -.62** | .63**  | .61**  | .47**  |        | .62**  | .72**  | .54**  | .58**  | .24** | -.06   | -.30** | .50**  | .20**  | .33**  | 4.4      | 1.0       |
| 9. Self-Concept         | 4.4      | 1.1       | .88**  | .54**  | .74**  | -.67** | .72**  | .59**  | .50**  | .69**  |        | .62**  | .36**  | .50**  | .23** | -.10   | -.27** | .55**  | .37**  | .52**  | 4.5      | 1.0       |
| 10. Inviting Self       | 4.4      | 1.1       | .54**  | .62**  | .58**  | -.50** | .56**  | .49**  | .36**  | .71**  | .56**  |        | .65**  | .58**  | .22** | .03    | -.14*  | .54**  | .13*   | .29**  | 4.3      | 1.1       |
| 11. Inviting Others     | 4.7      | 0.9       | .33**  | .45**  | .36**  | -.30** | .33**  | .35**  | .22**  | .47**  | .30**  | .55**  |        | .46**  | .18** | .02    | -.18** | .37**  | .14*   | .23**  | 4.3      | 1.1       |
| 12. Task Goals          | 3.9      | 1.2       | .45**  | .72**  | .51**  | -.45** | .43**  | .40**  | .32**  | .69**  | .48**  | .64**  | .45**  |        | .29** | .08    | -.11*  | .51**  | .02    | .14*   | 3.8      | 1.3       |
| 13. Approach Goals      | 4.1      | 1.2       | .05    | .32**  | .13*   | .11*   | .04    | .12*   | .10*   | .09    | .03    | .16*   | .15*   | .19**  |       | .45**  | .01    | .12*   | .05    | .08    | 4.3      | 1.1       |
| 14. Avoidance Goals     | 3.0      | 1.2       | -.33** | -.04   | -.22** | .44**  | -.28** | -.24** | -.13*  | -.28** | -.34** | -.21** | -.07   | -.13** | .44** |        | .28**  | -.18** | -.24** | -.17*  | 3.1      | 1.2       |
| 15. Self-Handicapping   | 2.5      | 1.2       | -.30** | -.22** | -.19** | .33**  | -.27** | -.29** | -.18** | -.34** | -.31** | -.27** | -.25** | -.25** | -.02  | .22**  |        | -.21*  | -.20** | -.26** | 2.7      | 1.2       |
| 16. Optimism            | 4.3      | 0.9       | .43**  | .42**  | .45**  | -.48** | .43**  | .45**  | .34**  | .56**  | .45**  | .52**  | .36**  | .45**  | .02   | -.31** | -.24** |        | .16*   | .27**  | 4.2      | 0.9       |
| 17. Teacher Rating      | 7.3      | 1.9       | .52**  | .09    | .38**  | -.37** | .45**  | .41**  | .29**  | .23**  | .45**  | .14*   | .18**  | .09    | .02   | -.25** | -.21** | .23*   |        | .67**  | 7.4      | 1.9       |
| 18. Semester GPA        | 86.0     | 8.3       | .69**  | .17**  | .46**  | -.41** | .60**  | .42**  | .34**  | .33**  | .61**  | .23**  | .24**  | .14*   | -.05  | -.28** | -.29** | .24**  | .72**  |        | 84.8     | 8.1       |

*Note.* Correlations for the girls ( $n = 408$ ) appear below the diagonal; correlations for boys students ( $n = 395$ ) appear above the diagonal. For calculations with Semester GPA,  $n$  for Girls = 398,  $n$  for Boys = 384. SE = self-efficacy.

\*  $p < .05$ . \*\*  $p < .001$ .

# Ch. 22 Overview

- Use and Interpretation of Multiple Regression
  - Correlation and Bivariate Regression
  - Conditions and Assumptions
  - Computing Multiple Regression (partial correlations)
  - **Hierarchical Multiple Regression**
  - **Simultaneous Multiple Regression**
  - **Stepwise Multiple Regression**
  - Logistic Regression
  - Discriminant Analysis

# What is Regression?

In linear regression, the regression coefficient represents the amount of change in the dependent variable for one-unit change in the independent variable.

# Defining Types of Regression

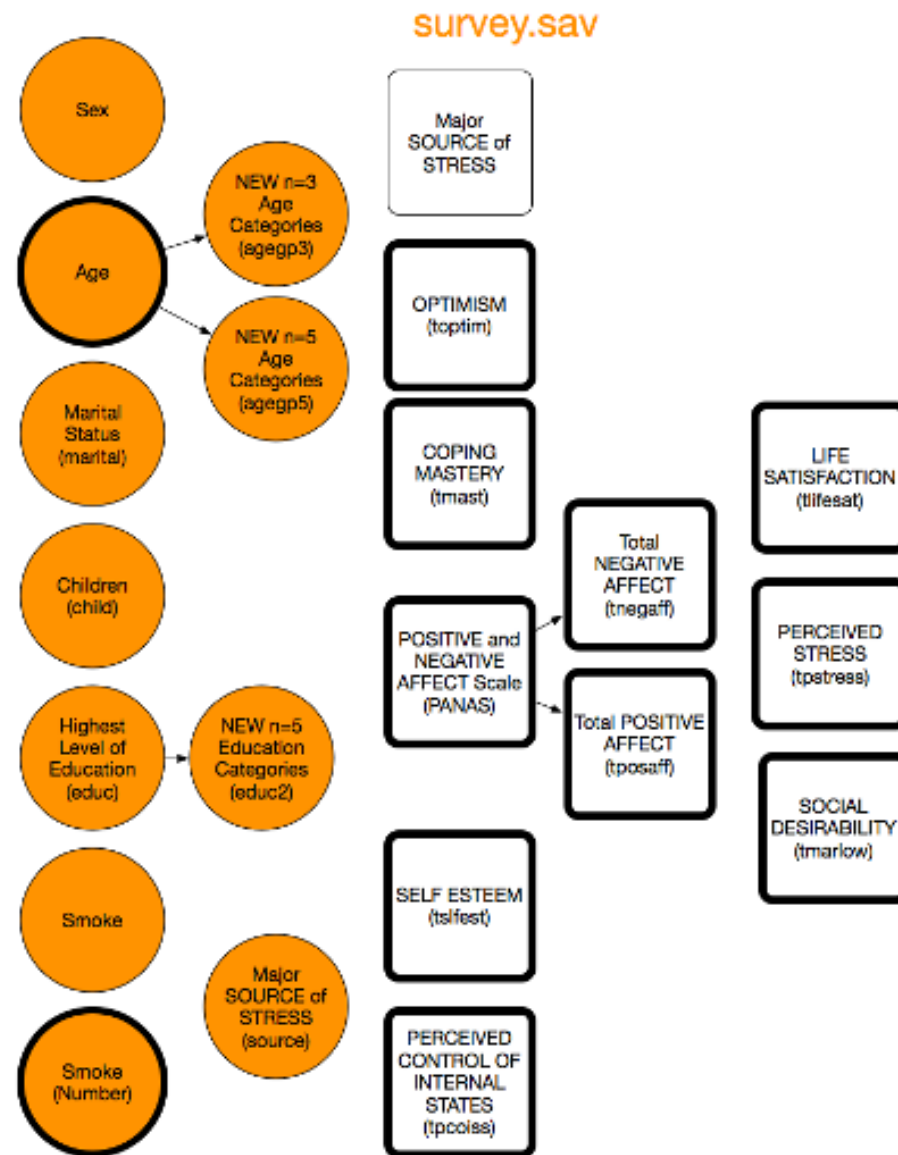
**TABLE 22.5**

Types of Regressions and Their Description

| Type of regression      | Description                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bivariate regression    | Regression analysis where one continuous variable predicts another continuous variable.                                                                                                                                                                                                                                                                        |
| Multiple regression     | A statistical method for analyzing data when there are several continuous and/or dichotomous independent variables and one continuous dependent variable.                                                                                                                                                                                                      |
| Hierarchical regression | A type of multiple regression that shows how variable(s)/ predictor(s) entered in a specific order improves the prediction.                                                                                                                                                                                                                                    |
| Stepwise regression     | A procedure that is similar to hierarchical multiple regression, but the computer instead of the researcher decides the order and how many of the potential predictors are used. The stepwise regression procedure describes how much more each independent or predictor variable has contributed to the prediction from the predictor variables already used. |
| Mediation               | A type of regression analysis where a variable (i.e., the mediating variable) reduces the magnitude of the relationship between two other variables.                                                                                                                                                                                                           |
| Moderation              | A type of regression analysis when the relationship between two variables is different depending on the level of a third variable.                                                                                                                                                                                                                             |
| Logistic regression     | A type of regression analysis where a categorical outcome variable is something that is predicted to happen or not.                                                                                                                                                                                                                                            |
| Discriminant analysis   | When there are several continuous predictor variables and a categorical outcome variable and when the goal is to categorize individuals or cases into categories.                                                                                                                                                                                              |



# Conducting the Tests in SPSS



# Correlation

Overview

Focus

Steps

Alternative?

SPSS

Also known as Pearson's Product Moment or Pearson's  
Assess relationship between *two continuous variables*

# Understanding the Output

## Correlations

|                                 |                     | tpstress Total perceived stress | tpcoiss Total PCOISS |
|---------------------------------|---------------------|---------------------------------|----------------------|
| tpstress Total perceived stress | Pearson Correlation | 1                               | -.581**              |
|                                 | Sig. (2-tailed)     |                                 | .000                 |
|                                 | N                   | 433                             | 426                  |
| tpcoiss Total PCOISS            | Pearson Correlation | -.581**                         | 1                    |
|                                 | Sig. (2-tailed)     | .000                            |                      |
|                                 | N                   | 426                             | 430                  |

\*\* . Correlation is significant at the 0.01 level (2-tailed).

## Nonparametric Correlations

|                |                         | tpstress Total perceived stress | tpcoiss Total PCOISS |
|----------------|-------------------------|---------------------------------|----------------------|
| Spearman's rho | Correlation Coefficient | 1.000                           | -.556**              |
|                | Sig. (2-tailed)         |                                 | .000                 |
|                | N                       | 433                             | 426                  |
|                | Correlation Coefficient | -.556**                         | 1.000                |
|                | Sig. (2-tailed)         | .000                            |                      |
|                | N                       | 426                             | 430                  |

\*\* . Correlation is significant at the 0.01 level (2-tailed).

# Effect Sizes for $r$

- Small  $r = .10$  to  $.29$
- Medium  $r = .30$  to  $.49$
- Large  $r = .50$  to  $1.00$

Cohen (1988, pp. 79-81)

# Writing it Up

The relationship between perceived control of internal states (as measured by the PCOISS) and perceived stress (as measured by the Perceived Stress Scale) was investigated using a Pearson product-moment correlation coefficient. Preliminary analyses were performed to ensure no violation of the assumptions of normality and linearity. There was a strong negative correlation between the two variables,  $r = -.58$ ,  $n = 426$ ,  $p < .001$ , with higher levels of perceived control associated with lower levels of perceived stress.

# Partial Correlation

Overview

Focus

Steps

SPSS

A measure of the relationship between the independent variable and the criterion variable, keeping the other independent variables constant.

TL;DR: It has a control.

# Understanding the Output

| Control Variables                 |                                   | tpcoiss Total PCOISS    | tpstress Total perceived stress | tmarlow Total social desirability |
|-----------------------------------|-----------------------------------|-------------------------|---------------------------------|-----------------------------------|
| -none <sup>a</sup>                | tpcoiss Total PCOISS              | Correlation             | 1.000                           | .295                              |
|                                   |                                   | Significance (2-tailed) | .                               | .000                              |
|                                   |                                   | df                      | 0                               | 425                               |
|                                   | tpstress Total perceived stress   | Correlation             | -.581                           | -.228                             |
|                                   |                                   | Significance (2-tailed) | .000                            | .000                              |
|                                   |                                   | df                      | 424                             | 426                               |
|                                   | tmarlow Total social desirability | Correlation             | .295                            | 1.000                             |
|                                   |                                   | Significance (2-tailed) | .000                            | .                                 |
|                                   |                                   | df                      | 425                             | 0                                 |
|                                   | tmarlow Total social desirability | Correlation             | 1.000                           | -.552                             |
|                                   |                                   | Significance (2-tailed) | .000                            | .                                 |
|                                   |                                   | df                      | 423                             | 0                                 |
| tmarlow Total social desirability | tpstress Total perceived stress   | Correlation             | -.552                           | 1.000                             |
|                                   |                                   | Significance (2-tailed) | .000                            |                                   |
|                                   |                                   | df                      | 423                             | 0                                 |
|                                   |                                   |                         |                                 |                                   |

a. Cells contain zero-order (Pearson) correlations.

# Effect Sizes

- Small  $r = .10$  to  $.29$
- Medium  $r = .30$  to  $.49$
- Large  $r = .50$  to  $1.00$

Cohen (1988, pp. 79-81)



# Writing it Up

Partial correlation was used to explore the relationship between perceived control of internal states (as measured by the PCOISS) and perceived stress (measured by the Perceived Stress Scale) while controlling for scores on the Marlowe-Crowne Social Desirability Scale. Preliminary assessments were performed to ensure no violation of assumptions of normality and linearity. There was a strong, negative partial correlation between perceived control of internal states and perceived stress, controlling for social desirability,  $r = -.55$ ,  $n = 425$ ,  $p < .001$ , with higher levels of perceived control being associated with lower levels of perceived stress. An inspection of the zero-order correlation coefficient ( $r = -.58$ ) suggested that controlling for socially desirable responding had very little effect on the strength of the relationship between these two variables.

# Multiple Regression

Overview

Focus

Assumptions

SPSS

Includes several continuous and/or dichotomous IVs and one continuous DV

# Checking MR Assumptions

Multicollinearity

Normality

Multivariate Outliers

Coefficients<sup>a</sup>

| Model |                      | Unstandardized Coefficients |            | Standardized Coefficients | t       | Sig.  | 95.0% Confidence Interval for B |             | Correlations |         |       | Collinearity Statistics |       |
|-------|----------------------|-----------------------------|------------|---------------------------|---------|-------|---------------------------------|-------------|--------------|---------|-------|-------------------------|-------|
|       |                      | B                           | Std. Error | Beta                      |         |       | Lower Bound                     | Upper Bound | Zero-order   | Partial | Part  | Tolerance               | VIF   |
| 1     | (Constant)           | 50.971                      | 1.273      |                           | 40.035  | <.001 | 48.409                          | 53.474      |              |         |       |                         |       |
|       | tmasr Total Mastery  | -.625                       | .061       | -.424                     | -10.222 | <.001 | -.745                           | -.505       | -.612        | -.445   | -.362 | .729                    | 1.372 |
|       | tpcoiss Total PCOISS | -.175                       | .020       | -.360                     | -8.660  | <.001 | -.215                           | -.136       | -.581        | -.388   | -.307 | .729                    | 1.372 |

<sup>a</sup> Dependent Variable: tpstress Total perceived stress

# Interpreting the Output

Model Summary<sup>a</sup>

3

| Model | R                 | R Square | Adjusted R Square | Std. Error of the Estimate |
|-------|-------------------|----------|-------------------|----------------------------|
| 1     | .684 <sup>a</sup> | .468     | .466              | 4.274                      |

<sup>a</sup> Predictors: (Constant), tpcoiss Total PCOISS, tmast Total Mastery

<sup>b</sup> Dependent Variable: tpstress Total perceived stress

ANOVA<sup>a</sup>

| Model |            | Sum of Squares | df  | Mean Square | F       | Sig.               |
|-------|------------|----------------|-----|-------------|---------|--------------------|
| 1     | Regression | 6806.728       | 2   | 3403.364    | 186.341 | <.001 <sup>b</sup> |
|       | Residual   | 7725.756       | 423 | 18.264      |         |                    |
|       | Total      | 14532.484      | 425 |             |         |                    |

<sup>a</sup> Dependent Variable: tpstress Total perceived stress

<sup>b</sup> Predictors: (Constant), tpcoiss Total PCOISS, tmast Total Mastery

1

Coefficients<sup>a</sup>

2

| Model |                      | Unstandardized Coefficients |            | Standardized Coefficients | t       | Sig.  | 95.0% Confidence Interval for B |             | Correlations |         |       | Collinearity Statistics |       |
|-------|----------------------|-----------------------------|------------|---------------------------|---------|-------|---------------------------------|-------------|--------------|---------|-------|-------------------------|-------|
|       |                      | B                           | Std. Error | Beta                      |         |       | Lower Bound                     | Upper Bound | Zero-order   | Partial | Part  | Tolerance               | VIF   |
| 1     | (Constant)           | 50.971                      | 1.273      |                           | 40.035  | <.001 | 48.469                          | 53.474      |              |         |       |                         |       |
|       | tmast Total Mastery  | -.625                       | .061       | -.424                     | -10.222 | <.001 | -.745                           | -.505       | -.612        | -.445   | -.362 | .729                    | 1.372 |
|       | tpcoiss Total PCOISS | -.175                       | .020       | -.360                     | -8.660  | <.001 | -.215                           | -.136       | -.581        | -.388   | -.307 | .729                    | 1.372 |

<sup>a</sup> Dependent Variable: tpstress Total perceived stress

The results of the analyses presented above allow us to answer the two research questions posed at the beginning of this section. Our model, which includes control of external events (Mastery) and control of internal state (PCOISS), explains a statistically significant 46.8 per cent of the variance in perceived stress (Question 1). Of these two predictor variables, mastery makes the largest unique contribution (explaining 16 per cent of the variance in stress), although PCOISS also made a statistically significant contribution explaining 9 per cent of the variance, over and above that explained by the Mastery Scale (Question 2).

## Unstandardized Interpretation

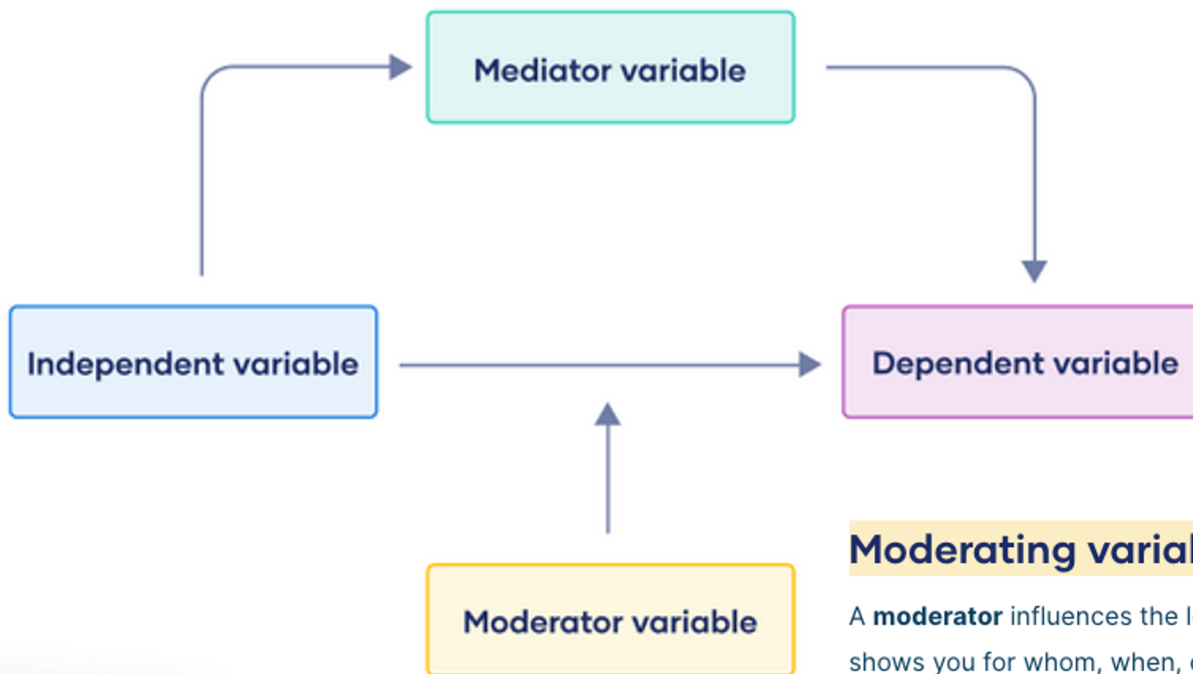
## Standardized Interpretation

For every 1 unit increase in mastery, perceived stress decreases by -.63 points

# Mediation & Moderation

## Mediating variables

A **mediator** is a way in which an independent variable impacts a dependent variable. It's part of the causal pathway of an effect, and it tells you how or why an effect takes place.



## Moderating variables

A **moderator** influences the level, direction, or presence of a relationship between variables. It shows you for whom, when, or under what circumstances a relationship will hold.

# Other Forms of Regression

Logistic Regression

Hierarchical Multiple Regression

Stepwise Multiple Regression

A categorical outcome variable is something that is predicted to happen or not. Based largely in probability as the outcome is dichotomous.

# Another Knowledge Check!

For questions 12–15, match the analysis with the particular question, and explain *why*.

- a. factorial ANOVA
- b. multiple regression
- c. factor analysis (refer back to Chapter 15)
- d. logistic regression

12. You have a pretest–posttest control group design. Your dependent variable is a 50-item questionnaire, which was given to 250 participants in the intervention group and 250 participants in the nonintervention group. Your next step is to reduce the number of questions to a smaller number of composite/summated variables.
13. You are interested in predicting whether people are successful or unsuccessful when they return to work. Your predictor variables are strength, range of motion, IQ, and gender.
14. You have two independent variables each with three levels, and you have one normally distributed dependent variable.
15. You are trying to predict GPA in graduate school in an English department. Your predictor variables are undergraduate GPA, Graduate Record Examination (GRE) verbal scores, score on an interest inventory, and age.



# Summary and Review

- Changing up next week to give more hands on analysis
- Example write ups and tables in Canvas
- There are too many types of regression
- Cannot say 'predict' unless design is experimental
- Theory is still driving choices

# Workshop: Connecting Surveys to CI SONA

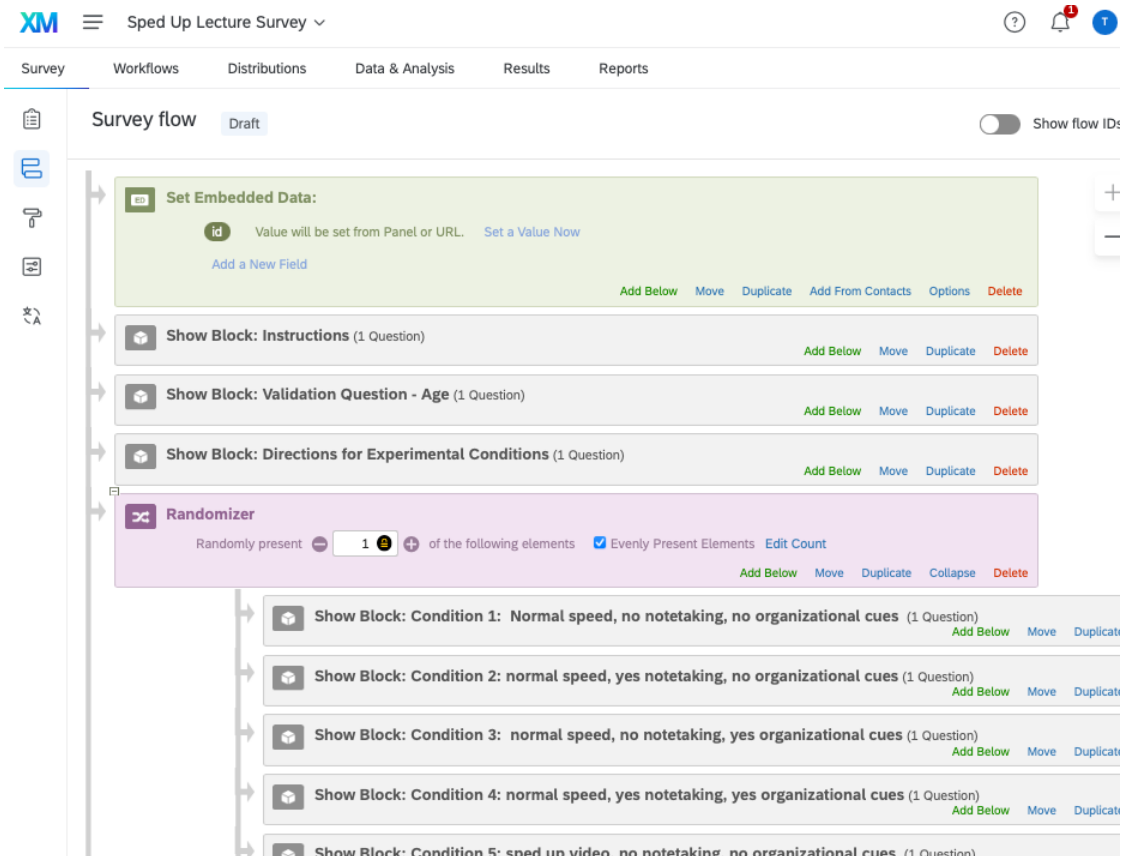
# Broad Steps

- Finish surveys
- Double check them
- Complete research subjects request form
- Get IRB approval and retain approval number
- Add survey to SONA

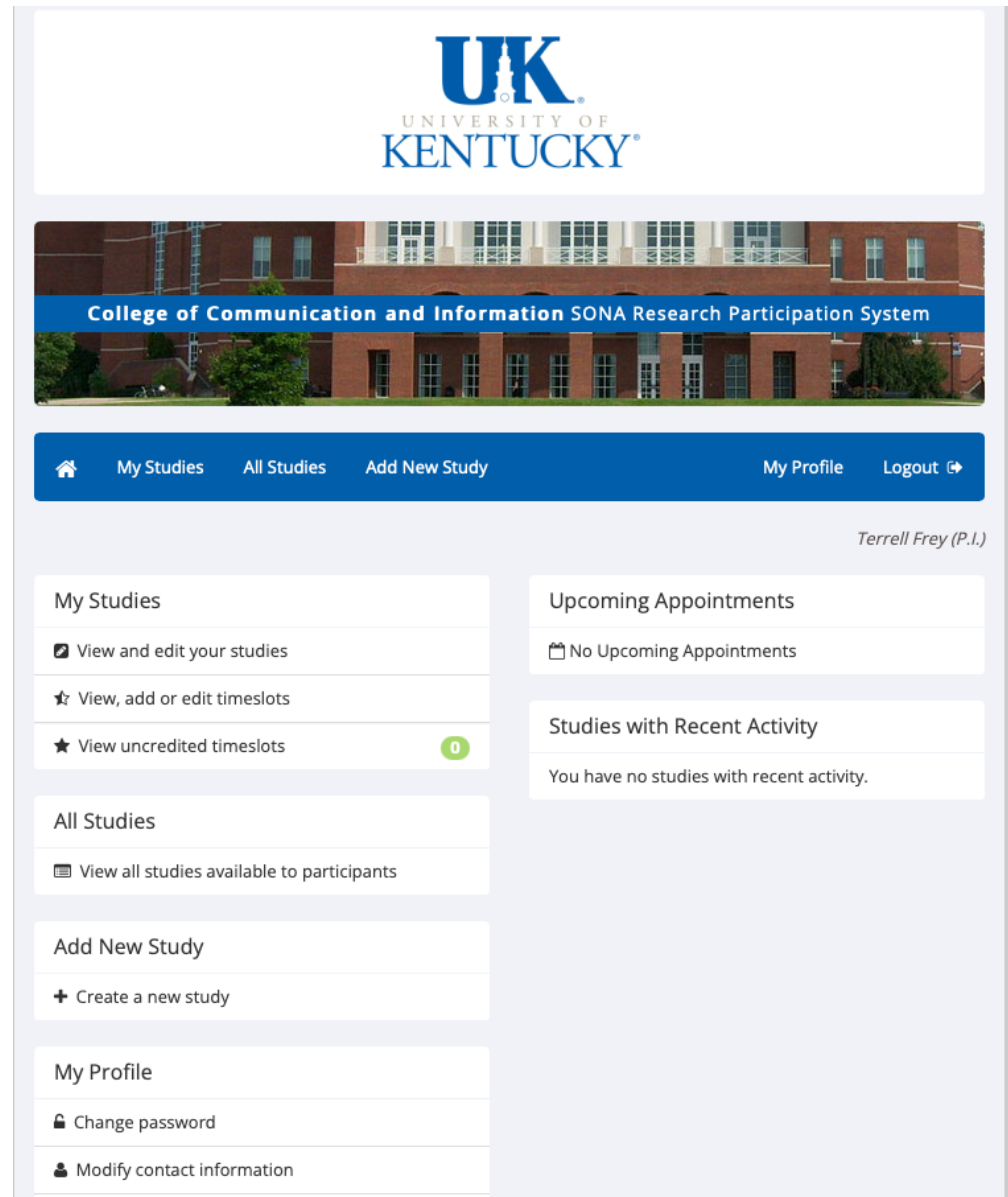
# Embedd ID

For Qualtrics:

- MUST add embed data to survey flow



# Setting up CI SONA



# Selecting the Type of Study

[Home](#) [My Studies](#) [All Studies](#) [Add New Study](#) [My Profile](#) [Logout](#)

Terrell Frey (P.I.)

Select Study Type

To add a new study, you must first select the type of study you would like to add. You may not change the study type once you have added it, so please be sure to select the correct option now.

 If you are conducting your study over **videoconference**, like Zoom, Webex, or a similar tool, set it up as either a  Standard Study or a  Multi-Part Standard Study.

☐ Standard Study



A study that is scheduled to take place at a specific time, in a specific place or over videoconference, and where there is only one part to the study that participants will participate in.

☐ Multi-Part Standard Study



A study that is scheduled to take place at a specific time, in a specific place or over videoconference, in multiple parts. The different parts may be scheduled to take place a specified number of days apart, and a participant must sign up for all parts of the study at one time.

☐  Two-Part

☐  Three-Part

☐  Four-Part

☒ Online External Study



An online study located on another website like Qualtrics, SurveyMonkey, Gorilla, PsychoPy (Pavlovia), REDCap, or any other website.

☐ Multi-Part Online External Study



An online study located on another website like Qualtrics, SurveyMonkey, Gorilla, PsychoPy (Pavlovia), REDCap, or any other website, in multiple parts. The different parts may be scheduled to take place a specified number of days apart, and a

# Adding the tag

Study URL

If the text **%SURVEY\_CODE%** is included in the URL, the system will replace that with a unique code for the participant, to make it easier to identify who completed the study.

[Detailed Help](#)

# Redirecting

After SONA generates your unique URL, you must add this back to Qualtrics so that participants are redirected



# Scott Johnson

Add Scott Johnson (scott.johnson@uky.edu) as a collaborator.

He is the resident Qualtrics guru and keeper of the SONA.